THE BASE CHANGE OF AN ETALE MORPHISM IS ETALE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. Same proof shape as this lane's already registered S1-morph-composition-etale (L14182), which quotes the same general lemma together with the same algebra input for the composition statement; the span names the mechanism and both inputs, so it is accepted as a proof and not treated as a bare pointer.

Allowed standard prerequisites (editorial).

- The general lemma turning a local ring-map property stable under base change into a base-change stable morphism property. Lemma \ref{lemma-composition-type-P}, cited at proof lines 8143-8146. Supplies the passage from rings to morphisms.
- Locality of the property 'the ring map is etale'. established in the proof of Lemma \ref{lemma-etale-characterize}, referred to at proof lines 8141-8143. Puts the property into the form required by the general lemma.
- Stability of the ring-map property under base change. Algebra, Lemma \ref{algebra-lemma-etale}, cited at proof lines 8146-8148. Verifies the hypothesis of the general lemma.

Source reference. Original lemma 02GO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3235–3235. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 02GJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3230–3230. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01SV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2352–2352. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00U2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1096–1096. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *The base change of a morphism which is étale is étale.*

Proof. In the proof of Lemma 02GJ we saw that being étale is a local property of ring maps. Hence the lemma follows from Lemma 01SV combined with the fact that being étale is a property of ring maps that is stable under base change, see Algebra, Lemma 00U2. □